

Here are study flashcards based on the content of **Sessions 6 & 7: Memory Management**:

Flashcard 1: Types of Memory

Q: What are the types of memory in a computer system?

A:

1. **Primary Memory (RAM):** Volatile, fast, used for current processes.
 2. **Secondary Memory:** Non-volatile, slower, long-term storage (e.g., HDD, SSD).
 3. **Cache Memory:** High-speed memory near the CPU for frequently accessed data.
 4. **Virtual Memory:** Logical extension of RAM on the hard disk.
-

Flashcard 2: Need for Memory Management

Q: Why is memory management essential?

A:

- Efficient allocation of memory.
 - Prevents wastage and fragmentation.
 - Ensures isolation between processes.
 - Allows concurrent execution.
-

Flashcard 3: Continuous vs. Dynamic Allocation

Q: What is the key difference between continuous and dynamic memory allocation?

A:

- **Continuous Allocation:** Fixed-size partitions; simple but prone to internal fragmentation.
 - **Dynamic Allocation:** Allocates memory at runtime; reduces fragmentation.
-

Flashcard 4: First Fit Allocation

Q: How does the First Fit memory allocation strategy work?

A: Allocates the first block large enough for the process; fast but can lead to fragmentation.

Flashcard 5: Best Fit Allocation

Q: What is the main advantage of Best Fit memory allocation?

A: Reduces wastage by selecting the smallest block that fits the process size.

Flashcard 6: Worst Fit Allocation

Q: What is the disadvantage of the Worst Fit memory allocation strategy?

A: It can waste large memory chunks.

Flashcard 7: Fragmentation

Q: What are the types of fragmentation in memory management?

A:

- **Internal Fragmentation:** Unused memory within allocated blocks.
 - **External Fragmentation:** Free memory exists but is non-contiguous.
-

Flashcard 8: Compaction

Q: What is the purpose of compaction in memory management?

A: Combines free memory into a single large block to reduce fragmentation.

Flashcard 9: Segmentation

Q: What is segmentation in memory management?

A: Divides processes into segments (e.g., code, data, stack) stored in different memory locations.

Flashcard 10: Paging

Q: How does paging eliminate external fragmentation?

A: By dividing memory into fixed-sized blocks (frames) and processes into pages of the same size.

Flashcard 11: Page Table

Q: What is the role of a page table in paging?

A: Maps logical page numbers to physical frame numbers.

Flashcard 12: Dirty Bit

Q: What does the dirty bit signify in paging?

A: Indicates if a page has been modified in memory but not written to disk.

Flashcard 13: Shared Pages

Q: How are shared pages used in memory management?

A: Multiple processes can access the same physical page, e.g., shared libraries.

Flashcard 14: Reentrant Code

Q: What is reentrant code?

A: Code that can be safely shared among multiple processes, like the code in `printf()`.

Flashcard 15: Throttling

Q: What is throttling in system resource management?

A: Limits resource usage by processes to prevent system overload.

Flashcard 16: I/O Management

Q: What are the key responsibilities of I/O management?

A:

- Allocating buffers.
 - Handling interrupts.
 - Managing device drivers.
-

Let me know if you need more cards or adjustments!